Lecture 20: Linear Transformations and Expectations of Multiple RVs

1. Definitions and Statistical Representations

- **Random Vector:** For random variables $X_1, X_2, \ldots, X_N$, they are represented as a vector denoted by $\underline{X}$. This provides a compact representation of multiple random variables, structured as $\underline{X} = [X_1, X_2, \ldots, X_N]^T$.
- **Mean Vector ($\underline{\mu}_X$):** The expected value of a random vector, defined as $\underline{\mu}_X = E[\underline{X}]$. It represents the average value of the random vector, where each element corresponds to the mean of one individual random variable.
- **Correlation Matrix (R_{XX}):** Defined as $R_{XX} = E[\underline{X}\underline{X}^T]$. Each element within the matrix is calculated as $R_{XX}(i,j) = E[X_i X_j]$. R_{XX} is a symmetric matrix.
- **Covariance Matrix (C_{XX}):** Acts as the "relationship map" of random vectors. The mathematical definition is $C_{XX} = E[(\underline{X} - \underline{\mu}_X)(\underline{X} - \underline{\mu}_X)^T]$. The matrix is symmetric, meaning $Cov(X_i, X_j) = Cov(X_j, X_i)$.
- **Relationship between Covariance and Correlation:** $C_{XX} = R_{XX} - \underline{\mu}_X \underline{\mu}_X^T$.

2. Linear Transformations

- **The Linear Model:** Consider the transformation $\underline{Y} = \underline{AX} + \underline{b}$.
  - $\underline{A}$ is the transformation matrix (scales and rotates the data).
  - $\underline{b}$ is the constant vector (shifts/translates the center).
  - $\underline{X}$ is the input random vector.
- **Transformation Properties:** If $\underline{Y} = \underline{AX} + \underline{b}$, the mapping properties are:
  - **Mean:** $\underline{\mu}_Y = \underline{A}\underline{\mu}_X + \underline{b}$.
  - **Covariance:** $C_{YY} = \underline{A} C_{XX} \underline{A}^T$.
  - **Correlation:** $R_{YY} = \underline{A} R_{XX} \underline{A}^T + \underline{A}\underline{\mu}_X \underline{b}^T + \underline{b}\underline{\mu}_X^T \underline{A}^T + \underline{b}\underline{b}^T$.

3. Multivariate Gaussian Distributions

- **1D Univariate Gaussian:** For a single RV X, the PDF is $f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$.
- **N-dimensional Multivariate Gaussian:** For a vector $\underline{X} = [X_1, X_2, \ldots, X_N]^T$, the general vector PDF is $f_{\underline{X}}(x) = \frac{1}{\sqrt{(2\pi)^N \det(C_{XX})}} \exp\left(-\frac{1}{2}(x - \underline{\mu}_X)^T C_{XX}^{-1} (x - \underline{\mu}_X)\right)$.
  - $\underline{\mu}_X$ is the mean vector ($N \times 1$).
  - C_{XX} is the covariance matrix ($N \times N$).

4. Derivations and Proofs

Derivation 1: Mean of Transformed Vector

- **Step 1:** Start with the definition $\underline{\mu}_Y = E[\underline{Y}]$.
- **Step 2:** Substitute the linear equation into the expectation: $E[\underline{AX} + \underline{b}]$.
- **Step 3:** Apply linearity of expectation to separate terms: $E[\underline{AX}] + E[\underline{b}]$.
- **Result:** $\underline{\mu}_Y = \underline{A}\underline{\mu}_X + \underline{b}$.

Derivation 2: Covariance Matrix Transformation

- **Step 1:** Observe that the deviation from the mean is independent of the shift $\underline{b}$: $\underline{Y} - \underline{\mu}_Y = (\underline{AX} + \underline{b}) - (\underline{A}\underline{\mu}_X + \underline{b}) = \underline{A}(\underline{X} - \underline{\mu}_X)$.
- **Step 2:** Substitute this into the definition of covariance: $C_{YY} = E[(\underline{Y} - \underline{\mu}_Y)(\underline{Y} - \underline{\mu}_Y)^T]$.
- **Step 3:** Expand the terms: $C_{YY} = E[\underline{A}(\underline{X} - \underline{\mu}_X)(\underline{X} - \underline{\mu}_X)^T \underline{A}^T]$.
- **Result:** Simplified to $C_{YY} = \underline{A} C_{XX} \underline{A}^T$.

Proof: Uncorrelated Gaussian Random Variables

- **Statement:** The joint Gaussian PDF for a vector of uncorrelated random variables factorizes into the product of individual Gaussian PDFs.
- **Step 1:** For mutually uncorrelated Gaussian random variables, $Cov(X_i, X_j) = 0$ for all $i \neq j$.
- **Step 2:** The covariance matrix C_{XX} simplifies to a diagonal matrix, meaning its inverse C_{XX}^{-1} is also diagonal with elements σ_n^{-2}.
- **Step 3:** Substituting the inverse back into the density function exponent yields $\exp\left(-\frac{1}{2}\sum_{n=1}^{N}\left(\frac{x_n-\mu_n}{\sigma_n}\right)^2\right)$.
- **Result:** The joint PDF factorizes into $\prod_{n=1}^{N}\frac{1}{\sqrt{2\pi\sigma_n^2}}\exp\left(-\frac{(x_n-\mu_n)^2}{2\sigma_n^2}\right)$.

5. Worked Examples

Example 1: Numerical Linear Transformation

- **Problem:** Given $\underline{Y} = \underline{AX}$ where $\underline{\mu}_X = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$, $\underline{A} = \begin{bmatrix} 1 & 0 & -1 \\ 0 & -1 & 1 \\ -1 & 1 & 0 \end{bmatrix}$, and $R_{XX} = \begin{bmatrix} 13 & 2 & 3 \\ 2 & 10 & 3 \\ 3 & 3 & 10 \end{bmatrix}$. Calculate the mean vector, correlation matrix, and covariance matrix of $\underline{Y}$.

- **Solution (Mean Vector):** Using $\underline{\mu}_Y = \underline{A}\underline{\mu}_X$, we multiply the matrices to get $\underline{\mu}_Y = \begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix}$.

- **Solution (Correlation Matrix):** Using $R_{YY} = \underline{A}R_{XX}\underline{A}^T$. First, compute $\underline{A}R_{XX} = \begin{bmatrix} 10 & -1 & -7 \\ 1 & -7 & 7 \\ -11 & 8 & 0 \end{bmatrix}$. Next, multiply by $\underline{A}^T$ to get $R_{YY} = \begin{bmatrix} 17 & -6 & -11 \\ -6 & 14 & -8 \\ -11 & -8 & 19 \end{bmatrix}$.

- **Solution (Covariance Matrix):** Using $C_{YY} = R_{YY} - \underline{\mu}_Y \underline{\mu}_Y^T$. Find the outer product $\underline{\mu}_Y \underline{\mu}_Y^T = \begin{bmatrix} 1 & 2 & -3 \\ 2 & 4 & -6 \\ -3 & -6 & 9 \end{bmatrix}$. Subtract this from R_{YY} to find $C_{YY} = \begin{bmatrix} 16 & -8 & -8 \\ -8 & 10 & -2 \\ -8 & -2 & 10 \end{bmatrix}$.

Example 2: Smart City Traffic Case Study

- **Scenario:** Designing a system in Ahmedabad predicting traffic congestion using transportation and climate variables.

- **Definition:** Define a random vector $X = [X_1, X_2, X_3, X_4, X_5]^T$, representing Traffic flow, Average vehicle speed, Rainfall intensity, Temperature, and Air pollution index (AQI).

- **Transformation:** The city defines a Congestion Index as a linear transformation $Y = 0.6X_1 - 0.4X_2 + 0.8X_3$.

- **Inference:** Assuming $X \sim \mathcal{N}(\mu, \Sigma)$, because $Y = a^T X$ is a linear transformation, Y is also Gaussian.